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Machine Learning for Predicting Success of Video Games

MASTER'S THESIS

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This is where a copy of the official signed thesis assignment and a copy of the Statement of an Author is located in the printed version of the document.

Declaration

Hereby I declare that this paper is my original authorial work, which I have worked out on my own. All sources, references, and literature used or excerpted during elaboration of this work are properly cited and listed in complete reference to the due source.

Michal Trněný

Advisor: Lubomír Popelínský

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Abstract

This thesis explores the subject of video games' success prediction. Existing attempts are discussed as well as factors traditionally believed to affect sales. A database of games released on the PC platform is created and the data is graphically presented. Lastly, experiments utilizing machine learning methods are conducted with the aim of discovering how well post-release success of PC games can be predicted before their release.

Keywords

video games, data mining, machine learning, Steam

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1 Introduction

The video games market has seen a large growth since its inception in 1970s to the point where video games have become a daily form of entertainment for many people of all ages around the world. It is a highly profitable market, reaching \$16.5 billion in U.S. sales in 2015 [2]. In comparison, the movie industry sold \$29.2 billion in 2015 in the U.S. [3][4]

The PC games market has seen an increase in digital sales and the number of releases after Valve launched its Steam Store¹. This store saw a major growth after 2012 thanks to a program called Greenlight, allowing developers to relatively easily release their games on Steam without a publisher which had been very difficult until then[5].

Over time, Greenlight led to a massive increase in the number of releases on the store, reaching over 10,000 by the end of 2016[6]. As a result, it became increasingly difficult for developers to stand out and even sell enough copies to fund the development of their games. If the success of games could be predicted beforehand, it would allow game creators to adjust the development to attract a larger audience or perhaps forsake their attempt to develop a game if it did not lead to a success. Thus, it would be useful to evaluate a concept; not an already finished game.

Previous attempts at video games success prediction assumed the game was already released and made predictions based on that knowledge. In addition, they dealt mostly with pre-2010 console games when there was no incentive to study the PC games market.

The goal of this thesis is to study known factors affecting the success of video games, create a database of PC games as no suitable one is available and, finally, estimate a game's success based on descriptive information such as genre, price, developer, or game features.

Chapter 2 presents applications of machine learning in the field of video games' success prediction. Chapter 3 describes what is currently believed to influence a game's post-launch success. The process of downloading the data required for this study can be found in Chapter 4 and its preparation in Chapter 5. Visual description of the data can

1. <http://store.steampowered.com/>

be seen in Chapter 6. Finally, conducted experiments and their results are presented in Chapter 7.

2 Related Work

This chapter presents papers and studies related to the topic of video games' success prediction.

2.1 Conference Papers

IEEE Conference on Computational Intelligence and Games¹ stands out as a prominent source of papers dealing with applying machine learning methods in video games development. The most common topics include automatic content generation and agent planning. While the topic of predicting revenue is not present, there are papers utilizing machine learning methods to predict factors related to games' success, such as retention or player experience.

2.1.1 Predicting Retention in Sandbox Games with Tensor Factorization-based Representation Learning

The authors processed data about at what times players were playing and what they were doing within the game. The goal was to learn from 14 days of activity and predict if the players keep playing after the following 7 days. The study is heavily focused on spatio-temporal data, i.e. how players travel within the game and how to process this data. Ensemble methods were mostly used for evaluation, achieving precision of 81 % and recall of 75 % in the best case.[7]

2.1.2 Predicting Player Churn in Destiny: A Hidden Markov Models Approach to Predicting Player Departure in a Major Online Game

The authors of this paper used very detailed data about player activities in a major title, Destiny. The data span across 17 months and included the activities of 10,000 players. Similarly to the previous study, the goal was to predict whether a player quits the game after a certain time window, in this case 4 weeks. They focused on the use of multinomial Hidden Markov Model which returned the highest

1. <http://www.ieee-cig.org/>

precision of 92 % with a relatively low recall of 43 % compared to other models.[8]

2.1.3 Transfer Learning for Cross-Game Prediction of Player Experience

The paper describes learning of how players experience one game and making predictions about thier experience in another game. The authors used statistical summarization of what players were doing and how well they were performing in two games. Players were then asked about their experience, namely engagement, frustration and challenge. The authors used two methods for the task of automatically mapping features between games, referred to as "supervised feature mapping" and "unsupervised transfer learning". Both methods produced accuracies above 58 % and 55 %, respectively, achieving 83 % accuracy on one of the subtasks (predicting challenge). These results were comparable with manual mappings created by experts.[9]

2.2 Other Studies

There are some studies, typically conducted by university students, which attempt to predict sales figures. However, they often lack a proper description of the data or results. Nevertheless, these are, to our best knowledge, the only publicly available studies on the topic of sales prediction.

2.2.1 Predicting Video Game Sales in the European Market

This study focused on game and console sales in Europe from March 12, 2005 to December 31, 2011. The authors used data about 2,450 games. The dataset contained 9 attributes and sales which they were attempting to predict. Simple regression models were fitted to predict weekly sales based on the first 2-6 weeks of sales. A prediction method for total sales was manually crafted and tested on all the data.[10]

2.2.2 Predicting Video Game Sales Using an Analysis of Internet Message Board Discussions

The aim of this thesis was to collect gaming forum posts and use this data to predict sales of video games. The data was collected from 2008 and 2009 from a major gaming message board. The author extracted mentions of each game and used the number of these mentions as well as sales from previous two weeks to predict sales in the upcoming weeks. The only evaluation metric used is Mean Absolute Error, making any conclusion of the results difficult.[11]

2.2.3 The Game Prophet: Predicting the success of Video Games

This study was using data about US, Japan, and European sales from 2001-2008. There were six attributes and "site hits", not further specified. The sale numbers were divided into 6 classes. The accuracy was mostly around 50 %, ranging from 70 % to 85 % based on the region when allowing a deviation by 1 class.[12]

2.3 Discussion

All known studies focusing on prediction of sales use data from VGChartz². The site does not clearly state the origin of the data nor what kind of sales exactly they are tracking[13]. Even if sales for consoles were accurate, the sales of PC games are clearly inaccurate. Some of the largest hits in the recent years, *Stardew Valley*³ and *Undertale*⁴, sold no units whatsoever according to the site. Examining the database suggests that VGChartz does not track digital sales, making it a completely unsuitable source of games' success on the PC platform.

We can only speculate about why there are no papers properly investigating sales prediction. There is a clear issue with the lack of accurate sales figures as publisher do not want to release exact numbers. This is understandable as it could, for instance, negatively affect stock prices. It is worth noting that the data used in the papers from the IEEE CIG conferences are directly provided by publishers

2. <http://www.vgchartz.com/>

3. <http://www.vgchartz.com/gamedb/?name=stardew+valley>

4. <http://www.vgchartz.com/gamedb/?name=undertale>

and employees of these publishers are involved in these studies. A study on revenue prediction would have to be conducted with the help from a publisher. Understandably, this publisher would like to keep any results to themselves as revenue would be directly involved. Not only not to reveal any precise numbers but also to use these findings to gain an advantage over their competition.

3 Known Factors Influencing Games' Success

With increasing competition on the video games market, those involved in games development and publishing keep speculating about what factors influence a game's success. Positive reviews in magazines used to be considered a key factor in what drives sales but with the rise of video-on-demand and real-time streaming services such as YouTube and Twitch, an increasing number of potential buyers can be seen turning away from traditional reviews towards video content. This chapter will present some findings on this topic from the recent years.

3.1 Clicks in Search Engine Results

In its white paper, "Understanding the Modern Gamer"[14], Google analyzed searches and clicks related to games from the years of 2010 and 2011. They focused on both desktop and mobile games. The first part of the study shows how the number of searches increased within a year and shows what gamers searched for pre-launch, during release and in the following months. Only 25 best-selling titles in 2011 were further analyzed. The study revealed 0.92 correlation between "clicks during the 10 month game launch cycle and game units sold during the first 4 months post-release". The study concluded that clicks from searches are a good predictor, however, "other factors – such as game quality, TV investment, online display investment, social buzz, and more" need to be considered to receive better predictions.

3.2 Reviews

Ars Technica came up with a way to estimate sales on Steam which they compared with review scores from Metacritic for around 1,300 games. Metacritic, among other forms of media, aggregates video game review scores from various websites and uses its own weighting to calculate an average score for each game. The findings, described in an article[15], show that reviews somewhat correlate with sales but there are too many outliers. One interesting highlight is that games

with score of 90 and above have significantly higher minimum sales than other games with lower scores. Games with scores below 60, on the other hand, never sold more than 1 million copies. They concluded that while games with higher score generally sell better, there are other, more important factors influencing sales of most games.

3.3 Video Services

YouTube is a popular service among gamers who produce and/or consume video content revolving around video games, including reviews, first impression, or gameplay. Many of these content producers have enough revenue from advertisements to create this form of content as a full-time job.

Sergey Galyonkin is mostly known for running the site Steam Spy¹, which was inspired by the previously mentioned Ars Technica's way of measuring sales on Steam. In his post[16], he showed a noticeable correlation between YouTube views and sales in the first month after a game's release. Further investigations revealed that the effect of YouTube may be long-term as games go on sale and potential customers search for information about these games even months after release.

Twitch is another notable video service, founded in 2011, which quickly became a popular streaming platform for gamers who can share their gameplay with anyone in the world in real time[17]. Similarly to YouTube, Twitch has become a source of sustainable income for many of those who appealed to a large audience.

Twitch conducted a study on the connection between Twitch views and sales on Steam, described in their blog post[18]. They claim that up to 25 % of sales come from users who watched a game on Twitch and bought it in the next 24 hours. They also show a 5% increase in retention for players who watched the given game in the same week they played it. However, this is based on 0.53 % of viewers that connected their Twitch account to Steam and may, therefore, be misleading.

While it is difficult to reliably link YouTube or Twitch views to sold copies, some clear examples can be occasionally seen. On March 8,

1. <https://steamspy.com/>

2017, the game Faeria² was released on Steam without having a large promotional campaign.

A day later, a popular games critic John Bain, known as "Total-Biscuit", was streaming the game on Twitch and released a video on YouTube three days later³. The game includes a recruitment system where both sides receive a reward if a player recommends the game to another person who buys it. John Bain was forced to stop playing the game as he received thousands of rewards, making the game unplayable. Steam Charts showed an increased interest in the game during these days as seen in Figure 3.1 which can be linked to John Bain thanks to the recruitment system.



Figure 3.1: Concurrent players of the game Faeria after release; provided by Steam Charts[19]

3.4 Discussion

Besides the sources mentioned above, there are other ways games can reach to their buyers, for instance social media, gaming communities and forums. However, games usually need to reach a certain level of quality in order to even have a chance to be noticed by some of the influencers. In other words, a game must be good. An unoriginal game

2. <http://store.steampowered.com/app/397060/Faeria/>

3. https://www.youtube.com/watch?v=GH6_X9otaiU

3. KNOWN FACTORS INFLUENCING GAMES' SUCCESS

with poor performance will hardly sell any copies. It is games that either do something new or do something well that are successful. Choosing e.g. the genre and setting determines future audience and consequently a sales potential. In this work, we explore how choosing the genre, theme, price, game features etc. influences games' success.

4 Data Acquisition

At the time of conducting this research, there was no publicly available machine-readable database of games released on Steam with extensive information about each game such as developer, publisher, price, genres etc. Steam Spy estimates the number of owners[20] (discussed later) but only lists few details about each game. Steam Charts tracks the number of concurrent players for every game[21] but nothing else. Steam Database[22] contains various information about each game such as prices, application packages, or concurrent players and owners (provided by Steam Spy). This data, however, is not complete nor available in a machine-readable form.

Therefore, we decided to create our own database, containing information about each game on Steam available before its release. This includes price, genres, developer, publisher, game features, languages etc. but excludes magazine reviews or user reviews.

In numerous cases, we downloaded this information years after a game's release. However, it is safe to assume that most information does not change after release, such as the genre, features etc. The price often changes, thus we found an alternative source listing price history. Small parts of a game's text description may also change but it only affects some games and not the whole texts. As a result, we assume that most of the information found on a game's Steam page did not change since its release.

4.1 Measuring Success

Valve does not provide sales figures for games released on Steam and they are not available anywhere else. Therefore, an alternative measure must be used. We downloaded data from two previously mentioned services: Steam Charts and Steam Spy.

Steam Charts collects data about concurrent players. For each game on Steam, the number of players currently playing the game is recorded every hour. The numbers are available since July, 2012 as monthly averages.

Steam Spy is a service that tracks the number of owners of each game, i.e. how many Steam users have each game in their library.

This, as stated on the website, includes "games bought on Steam, bought in retail and then activated on Steam, bought in bundles, received through promotions or as a gift and so on"[20]. Steam Spy was launched in April, 2015, significantly later than Steam Charts. Furthermore, the historical data is not accessible. Still, an estimate of sales seemed like a better option for measuring a game's success.

We obtained owner numbers from February, 2016 until the end of September, 2016 for roughly 10,000 games. The data was downloaded every day via the official API¹. A table containing the number of owners for each day was constructed. The data was missing for only several days and was therefore linearly extrapolated. We attempted to calculate sales, using this limited sample, for the rest of the data, described in Section 5.3.

Properly defining success is difficult and often depends on the expectations of the developer and the budget required to create a game. Selling one million copies can be a life-changing event for two friends working on small game that turned out to attract a lot of buyers. At the same time, it would be a tragedy for a team of 1,000 people working on an enormous project. There is, however, no way to determine the budget of games or their developers' expectations.

Thus, we will define success as the number of owners after two months since release for the data from Steam Spy, and the average number of concurrent players in two months after release for Steam Charts. Provided a developer would have plans on how many people will work on their upcoming game and how much it will cost, knowing this number would be enough to determine whether they end up in a profit or a loss or approximately how large the profit would be.

4.2 Download Process

The data was downloaded and processed using R scripts. Most of the required data was obtained from Steam itself.

1. <http://steamspy.com/api.php?request=all>

4.2.1 Steam API

There is a Steam API which is not officially documented[23] and provides almost complete information about games. One only needs to know the Steam ID of a game. A list of all IDs can be obtained at:

<http://api.steampowered.com/ISteamApps/GetAppList/v0001/>

However, it also includes all the DLCs (Downloadable Content), movies, and software, making the list unnecessarily long (around 40,000 entries). Steam Spy, on the other hand, has a list which excludes DLCs at:

<http://steamspy.com/api.php?request=all>

This list contains around 10,000 entries but still includes movies and software. These must be removed later.

The Steam API accepts multiple IDs but we decided to store each game's data in its own folder and hence downloaded each game's information separately. In addition, all Steam URLs support a country parameter which we used to fetch data from the U.S. version of the store in the following manner:

http://store.steampowered.com/api/appdetails?cc=us&appid= SOME_ID

This returns a JSON file containing most information normally visible on a store webpage. This includes the following data where the relevant entries will be discussed in Chapter 5:

- type
- name
- steam appid
- required age
- is free
- controller support
- detailed description
- about the game
- supported languages
- reviews
- header image
- website
- pc requirements
- mac requirements
- linux requirements

- developers
- publishers
- price overview
- packages
- package groups
- platforms
- categories
- genres
- screenshots
- movies
- achievements
- release date
- support info
- background

Some IDs lead to missing records which results in a JSON file beginning with `success = false` and these are skipped. Some games are stored under multiple IDs. To avoid duplicates, the currently processed ID is compared with the returned ID in the *steam_appid* attribute and the entry is skipped if the IDs do not match. Furthermore, an entry is skipped if it is marked as *coming_soon* or its type does not equal *game*.

The returned JSON file includes a list of links to a game's thumbnail and screenshots. Thumbnail is an image usually containing the game's name and is shown throughout the website whenever the game is listed, such as on a list of most popular releases, during sales etc. Screenshots are typically taken during gameplay and are displayed on a game's store page. These are downloaded to relevant folders.

4.2.2 Steam Website

Additional information is downloaded directly by parsing a game's store page, accessible via:

`http://store.steampowered.com/app/SOME_ID?cc=us`

This includes a release date which was found to be incomplete in some cases in the JSON files while being correctly displayed on the web page. The date is stored in three attributes: year, month, day.

Other data includes user tags which mostly represent detailed genre description entered by users. As Steam only lists 11 generic

genres, these can provide valuable additional information about a game. There is no limit on the number of user tags and these are, therefore, filtered later.

Age ratings were separately obtained from image names representing a specific age restriction as this information was often missing in the API output. The following ESRB (Entertainment Software Rating Board²) ratings are present on the U.S. version of the Steam Store:

- Rating Pending (RP) – rating not yet assigned
- Everyone (E) – no restrictions
- Early Childhood (EC) – 3+
- Teen (T) – 13+
- Mature (M) – 17+
- Adults Only (AO) – 18+

A store page includes a short description which is located at the top of a page and provides a brief description about the game. This is, however, not included in the JSON files and was extracted via *name="Description"* html tags.

There can be a DRM (Digital Rights Management) notice present, the common ones being:

- Account – the game requires setting up a 3-rd party account in order to play it.
- EULA – the user must agree to a 3-rd party license agreement before proceeding with an installation.
- SecuROM³ – a type of DRM limiting the number of installations with the goal of fighting piracy. SecuROM quickly became disliked by players who bought a legitimate copy but sometimes could not even install it.[24]

Steam has a program called Early Access which allows developers to release their game while in development, as long as it is playable[25]. It is commonly used to fund further development and/or collect feedback from players and direct development based on that.

These games follow a different release process than other games as there are essentially two releases – into Early Access and out of

2. <https://www.esrb.org/>

3. <https://www2.securom.com/>

Early Access. There may also be large updates during the Early Access process, possibly attracting new buyers. Such a game may have a slow start but gradually attract many buyers. This makes it difficult to evaluate their post-release success.

Hence, we decided to omit these games in further processing. Games currently in Early Access are labeled as Early Access. However, games that came out of Early Access no longer have this label. They do often have user reviews labeled "Early Access Reviews", though, found at:

http://steamcommunity.com/app/SOME_ID/reviews/?browsefilter=toprated&cc=us

Each game thus has a record of whether at least one of such reviews is present, indicating it used to be in the Early Access program.

4.2.3 Launch Price

Steam only lists current price which is often lowered as a game gets older. Steam Sales⁴ has sales history since around 2010 which is sufficient (due to Steam Charts data being available only since July, 2012). While the launch price is not listed, each price drop is accompanied by the previous price. It is safe to assume that the price before the first price drop was the original price of the game. The prices are extracted using a regular expression over html files from the URL:

http://steamsales.rhekua.com/view.php?steam_type=app&steam_id=SOME_ID

If a game has no record on Steam Sales and was released recently (in the last six months), it may be because it has not been on sale yet. In that case, the price on Steam can be assumed to be the original price.

4.2.4 Steam Charts

The data found on Steam Charts is stored in html tables at the URL:

http://steamcharts.com/app/SOME_ID.

The function *readHTMLTable* in the R package *XML*⁵ makes extraction of these tables trivial. Each downloaded table contains:

- Month – the year and the month each row represents

4. <http://steamsales.rhekua.com/>

5. <https://cran.r-project.org/web/packages/XML/>

- Avg. Players – the average number of concurrent players in the given month
- Gain – difference between current and previous month
- % Gain – the above as a percentage
- Peak Players – the maximum number of concurrent players recorded that month

4.3 Summary

Data about a total of 9,780 games was downloaded on October 6, 2016 and the process took around 13 hours, resulting in over 7 GB of data. In case of an error, there were 3 attempts for each ID with a 10 second timeout. In order not to exceed the API limits (which are not documented but present) and not to generate an excessive amount of traffic on any of the used websites, there was a 5 seconds waiting period before processing the next ID.

Each game has its own folder with the downloaded screenshots, a thumbnail, and an .RData file containing a nested list of the data from Steam API and the additional information described above. The next chapter describes the processing of this data.

5 Data Preparation

The next step is processing the separate .RData files, cleaning the data, removing unwanted entries, and building a single table. The result of this stage is two datasets: one containing most games on Steam – all the obtained games with at least partial information – which can be used to e.g. look up all the games from a certain developer. The second dataset contains all the necessary data and is used further for evaluation.

5.1 Raw Data Processing

Name, developer, publisher, short and long descriptions are cleaned off html tags which are occasionally present. In addition, name is converted to ASCII-only as it may contain redundant Unicode characters such as ®.

Games with a missing price value receive *NA* as their price and are marked as not relevant for further processing. As explained at the beginning of this chapter, these games are still included in the large dataset but are not present in the final dataset. The Steam API does not list a price if the game is free-to-play. These games rely on a different business model than premium titles (with an up-front price) as even a high number of players does not have to translate into a large revenue, making measuring their success difficult.

If a developer's or a publisher's name is empty, it is replaced with the word "unknown" + a unique number. This is the case in 21 and 28 cases, respectively, and thus does not represent a major issue.

Every game has a list of categories defined by Steam. These are either present or not present and include:

- Single-Player – the whole game or one of its modes revolve around interacting with the game environment or AI characters
- Multi-Player – the whole game or one of its modes revolve around interacting with other players, usually competing against each other
- Cross-Platform Multi-Player - Multi-player is not limited to PC only but players can be matched with players on other platforms, typically PlayStation or Xbox

- Co-op – game modes where players cooperate towards a common goal. There are only AI-controlled opponents
- Local Multi-Player – There are separate “Local Co-op” and “Local Multi-Player” categories. Both, however, refer to games typically meant for playing at home with a group of friends. Therefore, it makes sense to merge them into one
- Online Co-op – Co-op where players do not have to play on the same local network
- Split screen – a form of local multi-player where multiple players share one computer and one screen
- Steam Achievements – achievements are rewarded for certain actions in the game and displayed on a user’s profile
- Steam Leaderboards – the game shows players with the highest score in all or at least some of its game modes
- Steam Trading Cards – Steam has its own market with cards and other items rewarded for playing games. These can also be bought with a real-world currency, sold for credit (represented as a real-world currency), and used to buy games or other items. Badges displayed on a user’s profile can be crafted from these cards. A simple presence of Steam Trading Cards is believed to potentially boost a game’s sales.
- Steam Workshop – provides a user-friendly way of publishing and installing modifications for a game
- Level editor – another form of creating new content for a game by players
- Partial controller support – the game can be played with a controller as an alternative to mouse and keyboard but e.g. menu navigation may be difficult or even require a mouse/keyboard
- Full controller support – players should be able to perform all actions within the game, including menu navigation or inventory management, using a controller only.
- VR support – whether a game supports a Virtual Reality headset (HTC Vive or Oculus Rift)
- Steam Cloud – player’s progress is saved on Steam’s servers and can be synchronized across multiple devices
- Valve Anti Cheat – Valve’s software protection against cheaters
- Captions available – game includes captions over voice dialog

- In-App Purchases – game allows purchasing in-game currency or items, not common in premium titles

Each game was marked as “Early Access Now” if Early Access was one its categories during the data download. Games with reviews marked as Early Access received “Early Access Past” label. In both cases, such games are marked as not relevant for the final dataset.

The Steam API lists controller support as a separate attribute besides the Steam categories. However, the only values are “none” and “full”, while Steam makes a distinction between partial and full controller support, as seen in the previously mentioned categories. This attribute is therefore omitted.

The following game genres are present: RPG, Strategy, Adventure, Action, Simulation, Racing, Casual, Sports, Massively Multiplayer, Education, Indie. These were added as separate binary attributes.

The data also contains software that is not games, including: Animation & Modeling, Photo Editing, Utilities, Design & Illustration, Web Publishing, Audio Production, Software Training, Video Production, Accounting. These entries are completely removed from the dataset.

There is also a “Free-to-play” genre. As explained before, these games are marked as not relevant.

The number of screenshots and videos is counted. These allow potential buyers to see what the game looks like and, in the case of e.g. trailers, also its gameplay in motion. Games have 11 screenshots on average, 9 being the median. The minimum is 1 and maximum 114. There are 1.4 videos on average with 1 being the median. The minimum is 0 and maximum 20.

5.1.1 Hardware Requirements

Steam allows listing of both minimum and recommended hardware requirements for Windows, Linux, and Mac. We focus on Windows and minimum requirements only as recommended requirements are not always present and there is no clear definition of what recommended means, while minimum requirements state the kind of hardware a user must meet for the game to be playable, as determined by the developer. Hardware requirements do not follow any standardized

format and are simply stored as a string. We extracted the following five common features:

CPU

The required processing power can be found via the keyword "Processor", often in form:

Processor: Intel® Core™2 Duo 2.6 GHz or AMD Athlon™
64 X2 4000+ or better
Processor: 1 GHz Processor

We only look for the frequency using patterns such as "*Processor: [number] GHz | Gigahertz*". While higher frequency does not necessarily mean higher processing power and vice versa, the requirements do not always state the CPU model but only a frequency instead. This frequency was converted to MHz if necessary in order to obtain a unified format.

GPU

The required graphics card is listed after the keywords Graphics or Video Card. Examples:

Graphics: All NVIDIA® GeForce® 7800 GT 256 MB and
better chipsets. All ATI Radeon™ X1800 256 MB and
better chipsets
Graphics: NVIDIA GTX 970

In order to assign a numeric value to the model names, we used benchmark scores from Videocard Benchmarks¹. There are three charts from which model names and scores are extracted and merged into a single table containing 1,068 entries.

http://www.videocardbenchmark.net/high_end_gpus.html
http://www.videocardbenchmark.net/mid_range_gpus.html
http://www.videocardbenchmark.net/midlow_range_gpus.html

The string containing graphics requirements is matched against the model names in the table. If multiple matches are found, the lowest corresponding score is selected (since minimum requirements are concerned).

1. <http://www.videocardbenchmark.net>

RAM

RAM requirements can be extracted trivially as they are found in the form:

Memory: 256 MB RAM

Memory: 2 GB RAM

A simple pattern "*Memory: [number] MB | GB*" is enough to obtain a number which is converted to MB if necessary.

Storage

This states how much hard drive space a game takes and is displayed in the following way:

Storage: 1 GB available space

Hard Disk Space: 50 MB Hard Drive Space

The storage requirements can be obtained by looking for the pattern: "*Storage | Hard Disk Space | Hard Drive: [number] KB | MB | GB*".

DirectX

DirectX is a set of APIs (Application Programming Interface) most commonly used in the process of developing games for Windows operating systems[26]. Higher version, in conclusion, allows creating better-looking visuals. The version number can be trivially extracted by looking for a number:

DirectX: Version 9.0

DirectX: Version 12

There are a few obvious errors present such as *4000 GB RAM* which was clearly meant to be *4000 MB*. These can be simply corrected. If hardware requirements are not present or cannot be extracted, *NA* values are used instead.

5.1.2 Steam Charts

Steam Charts was the most important decisive factor in whether a game can be further processed or not as the data from this service

was intended to be used as the measure of success. Only games released since July 1, 2012, when Steam Charts was launched, are further processed.

Each table containing data from Steam Charts is checked for the presence of rows following the game's release, in case the relevant data is missing despite there being enough time between the game's release and data acquisition. The average number of concurrent players in two months following a game's release date is calculated from the table. The weight of each month is adjusted based on the number of days in each relevant month. For instance, if a game was released on January 21, the weights are 10 for January, 28 for February, and 21 for March.

5.2 Processing Relevant Entries

The dataset is ordered by year, month, and day. 9,568 games remain in the large dataset. Only relevant records are further processed which leaves 4,634 games in the dataset that is used in further processing. In addition, new attributes are inferred as described below.

While it makes sense to keep a release date as three separate attributes – year, month, day – it may be useful to present it in a timeline. Months are therefore counted since January, 2012 in one attribute. A game released in e.g. February, 2014 then receives the value 26. A weekday is further extracted from a release date using the *Weekdays* function in R.

Each game has a binary attribute stating whether it is a sequel to another game. This can be mostly determined from its name:

Far Cry 4

Grand Theft Auto V

TransOcean 2: Rivals

In addition, each game's description is searched for the words "sequel", "franchise", and "continuation" which likely point to the fact that the game is a sequel.

Our hand-crafted classifier based on regular expressions was tested on a random sample of 50 games, 10 of which were marked as a sequel. Three games were falsely marked as not a sequel while one game was falsely marked as a sequel based on manual research on these games.

This task can be complicated by games such as *Ravenword: Shadowlands*. The presence of a colon does not reliably mark a sequel and such games may thus be overlooked. They do, however, often have a mention about them being a sequel in their description. The one game that was falsely marked as a sequel had the word "sequel" in its description as a promotion of its future sequel.

An examination of games and their descriptions suggests that customization in some form (such as customizing a character's appearance) may be an interesting feature, not documented by other attributes. A binary attribute "Is customizable" is therefore added, stating whether one of the words "custom", "customize", or "customizable" is present in a game's description.

Each game is assigned information about games previously released by the same developer. The first attribute states how many games were released by its developer before this game.

The number of average concurrent players for each of these previous games is looked up, provided the number is available. These create a list of numeric values. The maximum and minimum of these values are noted. Lastly, their Gini index is calculated to represent whether a high maximum was due to one successful game or whether the developer keeps consistently making games of similar popularity.

The same is performed for publishers.

User tags, until now stored in a semi-colon-separated string are extracted into separate attributes. There are over 300 user tags present, hence most need to be filtered out. As a first step, we picked 5 most popular tags for each game if more were present. Examination of the tags shows that popular games may be assigned rare tags that are not completely relevant to the game but since these tags are sorted by popularity, the less relevant ones are located at the bottom of the list. Further, we only kept tags a developer would likely give their game during development. This meant mostly removing tags with a subjective context, such as: Memes, Cute, Great Soundtrack, Funny etc. The following tags are present in the dataset as separate attributes stating whether the tag was present or not:

2D, 4X, Board Game, Card Game, City Builder, Crafting, Cyberpunk, Dating Sim, Episodic, Family-Friendly, Fantasy, Female Protagonist, Fighting, First-Person, Flight, FPS, Hidden Object, Horror,

JRPG, Medieval, Noir, Nudity, Open World, Parkour, Pixel Graphics, Platformer, Point & Click, Puzzle, Remake, Retro, Rhythm, Roguelike, Rogue-lite, RTS, Sandbox, Sci-fi, Space, Stealth, Steampunk, Story-Rich, Superhero, Survival, Survival Horror, Third-Person, Third-Person Shooter, Tower Defense, Turn-Based, Turn-Based Strategy, Visual Novel, Walking Simulator, World War II, Zombies

Age requirements are converted to actual age or NA if undetermined.

- "e" (Everyone) <- 0
- "rp" (Rating Pending) <- NA
- "ec" (Early Childhood) <- 3
- "t" (Teen) <- 13
- "m" (Mature) <- 17
- "ao" (Adults Only) <- 18

There are 27 languages supported across all games on Steam. Only the 10 most frequent ones were selected and put into separate attributes, indicating whether the language is supported or not. Table 5.1 lists these languages along with the number of games they are supported by. In addition, the total number of supported languages is recorded for each game.

Table 5.1: Most frequently supported languages

Language	Frequency
English	4623
German	1628
French	1544
Spanish	1445
Italian	1231
Russian	1089
Japanese	671
Portuguese-Brazil	572
Polish	570
Simplified Chinese	353

The number of characters is counted for each game's name and both its long and short description. A short description typically provides a

brief overview of the game's genre and setting, while a long description contains detailed information about the game.

Images were processed separately as they were stored in the form of .jpg files. Only the thumbnail and the first screenshot that is displayed on a game's store page were processed as they are the first images a user sees from the game.

The number of distinct colors was counted. Colors were also reduced to a palette of 64 predetermined colors and the number of unique colors on this reduced image was noted. The intention was to detect images which consist of e.g. one color in various shades.

Colors of the original image, stored in RGB format, were converted into HSV (Hue-Saturation-Value) format. The hue values were divided into 16 bins, each bin representing the percentage of pixels belonging to this bin. An example image can be seen in Figure 5.1 and its hue values in table 5.2. Hue 1 and 16 refer to red colors, the green colors are mostly in Hue 5, and the dominant blue colors in Hue 9 and 10. In addition, the average saturation and value were calculated to detect, for instance, black and white images or highly colorful images.



Figure 5.1: thumbnail

Table 5.2: Example of an image's hue table

Hue1	Hue2	Hue3	Hue4	Hue5	Hue6	Hue7	Hue8
8.5	1.3	1.6	2.6	8.2	2.8	1.3	4.6
Hue9	Hue10	Hue11	Hue12	Hue13	Hue14	Hue15	Hue16
22.6	41	0.8	0.3	0.2	1.2	0.5	2.7

5.3 Comparing Steam Charts and Steam Spy

As described in Chapter 4, we separately downloaded data from Steam Spy which grants us information about the number of owners for each game. This mostly translates into sales accurately enough, however, some estimates have a margin of error as high as 38 % [27].

On the other hand, our metric based on the data from Steam Charts is not 100 % accurate, either, as the 2-month average is calculated from monthly averages and not, for instance, daily averages which would yield more accurate numbers. We compared these two metrics to see if the average number of concurrent players and the number of owners correlate enough to infer one from another.

The number of owners after two months since release was extracted to match the two-month period used for data from Steam Charts. This covered roughly 1,000 games present in the previously crafted dataset. The task was to determine how the number of owners and average concurrent players correlate and whether the number of average players could be used to derive the number of owners for games where this data was not available.

Binary logarithm was applied to both metrics to negate the effect of few highly successful games (described in the next chapter). Although we discovered a high correlation of 0.82 (Figure 5.2), it is not possible to derive the number of owners from average players with high enough accuracy.

The reason may be the inaccuracy of both measures and cases where users buy a bundle of several games, add them to their library but never play some of them. Other games may have a small, yet dedicated player base.

There is also a problem with games on the lower side of the spectrum. These may have a thousand owners but that is simply because the developer sent out keys to reviewers and gave out other keys for free as an attempt to establish at least a minimal player base. The data from Steam Charts says how this was successful. If a game had near zero players on average, then a thousand owners has may very well mean close no sold copies.

We concluded that the number of owners cannot be reliably derived from the number of average players and we cannot replace one metric with another as a consequence. Moreover, the number of av-

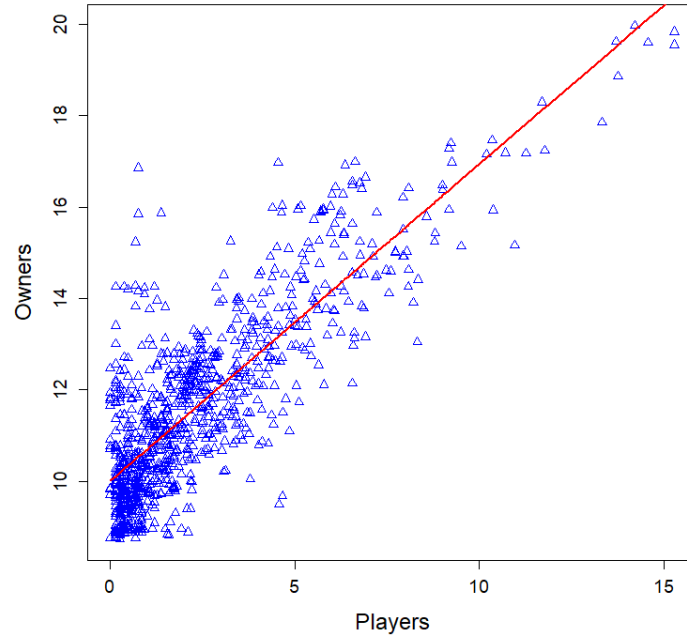


Figure 5.2: Dependence between average players and owners

erage players shows the actual interest in a given game compared to ownership. The data about average concurrent players is, therefore, used in the rest of this study.

6 Data Description

This chapter provides an overview of the data later used for evaluation and containing records about 4,634 games. The development of the number of monthly releases is shown as well as how the presence of a feature or the lack thereof correlates with a game’s success.

6.1 Development Throughout the Years

First, we show the number of releases across all the months covered in the dataset as seen in Figure 6.1. Valve launched Steam Greenlight in August 30, 2012, allowing developers to release their games on Steam without a publisher via community votes[5]. A year later, Valve made Greenlight less strict, allowing even more games onto the platform[28].

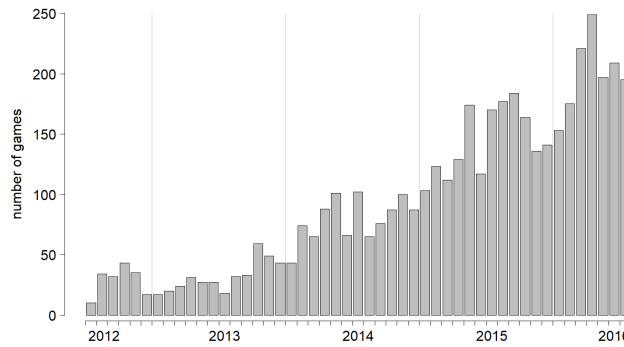


Figure 6.1: Games released on Steam and present in the dataset (August, 2012 – July, 2016)

Only a few games reach high popularity. A histogram of the average concurrent number of players can capture how many games averaged a certain number of players in the first two months after release. Figure 6.2 without any scale adjustments shows that all but few games had less than 10,000 players on average. There were only a few outliers with more than 10,000 players.

Removing these outliers and focusing only on games with less than 10,000 players would produce a similarly-looking graph, however. Thus, we introduce eight intervals by the average number of players:

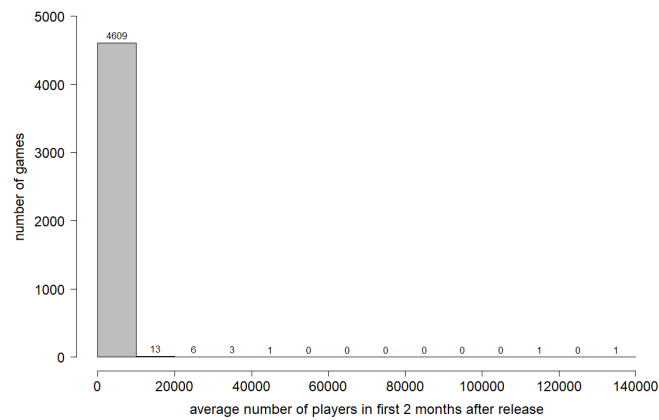


Figure 6.2: Average number of players in games (August, 2012 – July, 2016)

- 0-1
- 1-5
- 5-20
- 20-100
- 100-500
- 500-2,000
- 2,000-10,000
- 10,000-140,000

Figure 6.3 shows a similar histogram as Figure 6.2 but the intervals on the x-axis have been adjusted. This, for instance, shows that 33 % of all games had less than 1 player on average in the first two months.

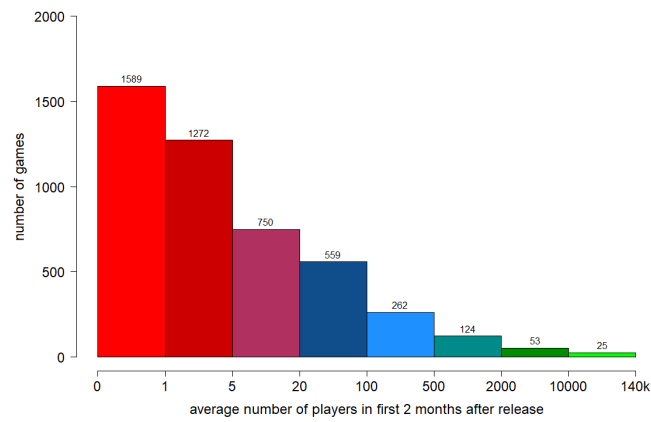


Figure 6.3: Average number of players in games (released August 2012 - July 2016)

The dataset contains complete data for the years 2013 – 2015. Figure 6.4 shows a comparison of these years: how many games were released in total and a rapid increase of less played games.

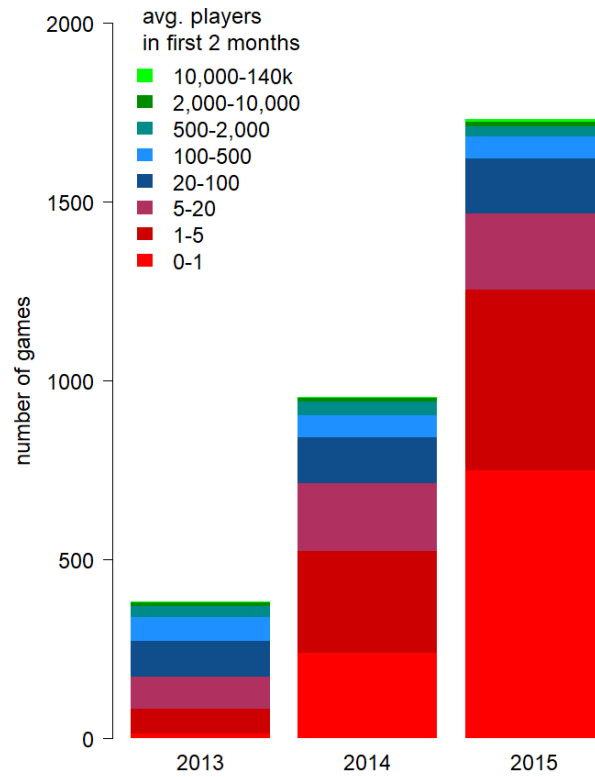


Figure 6.4: Games released on Steam 2013 - 2015

We will further focus on the year 2015 only as it provides the most recent available information across a whole year for a total of 1,730 games. The number of releases each month along with their average number of players after release is captured by Figure 6.5.

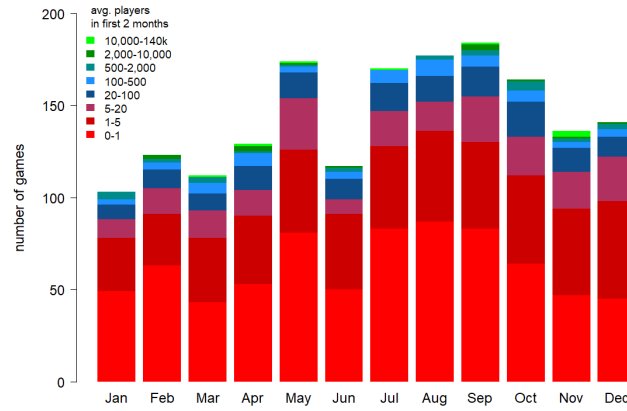


Figure 6.5: Number of games released in each month in 2015

6.2 Important Features

An overview of the more important features follows.

Price is an important factor for games. While most games cost \$10 and less, the more successful ones can be generally found above \$40 (Figure 6.6). This is caused by the fact that titles developed by large studios aim to make high-quality products and their budget allows them to invest into advertising, targeting a large audience willing to spend the money. Simply attaching a high price to a game is naturally not enough.

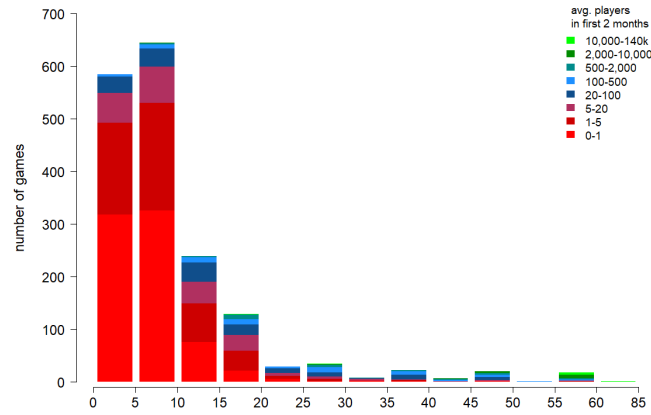


Figure 6.6: Price

Games with higher graphics and storage requirements are generally more successful as shown in Figure 6.7 and Figure 6.8. This, again, aligns with high-budged games investing into technology and creating visually appealing products.

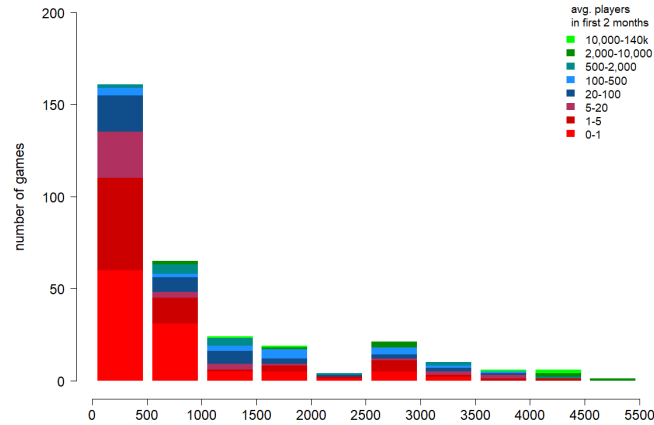


Figure 6.7: GPU benchmark scores (higher is better)

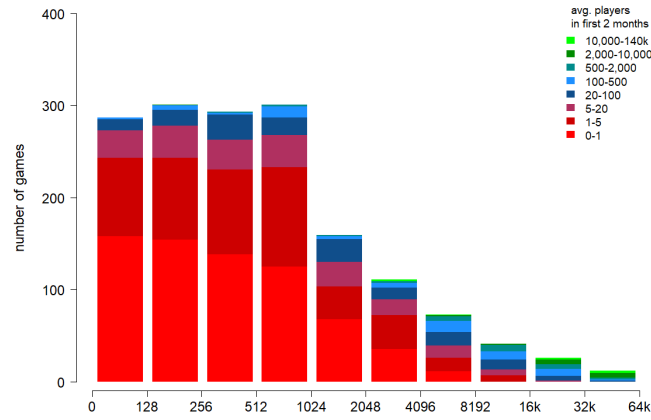


Figure 6.8: Storage requirements (MB)

Genres, as defined by Steam, are very generic. As a result, none of them stand out as a successful genre. That is why user tags were included in the data. We picked some of the more interesting user tags out of the 52 present in the dataset. Figure 6.9 shows games with tags which often reach a high number of average players. Open World and Third-Person (Shooter) especially stand out as highly successful. On the other hand, Figure 6.10 shows tags generally found attached to less successful games. This, for instance, shows that a 2D retro-inspired platformer is unlikely to attract a lot of attention.

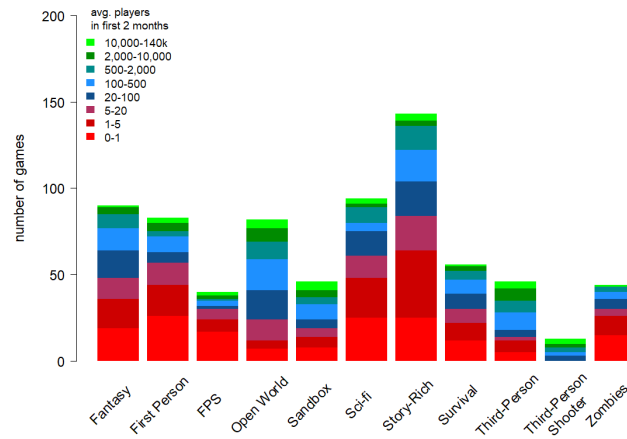


Figure 6.9: Successful tags

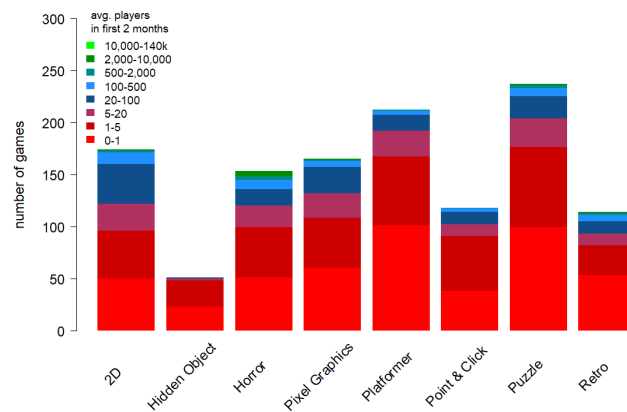


Figure 6.10: Unsuccessful tags

Multi-Player is generally considered an important feature but there have been very successful games without its support. Figure 6.11 shows games grouped up by the average number of concurrent players and how many from each group did and did not include multi-player.

Steam Achievements are almost a necessity in more successful games (Figure 6.12).

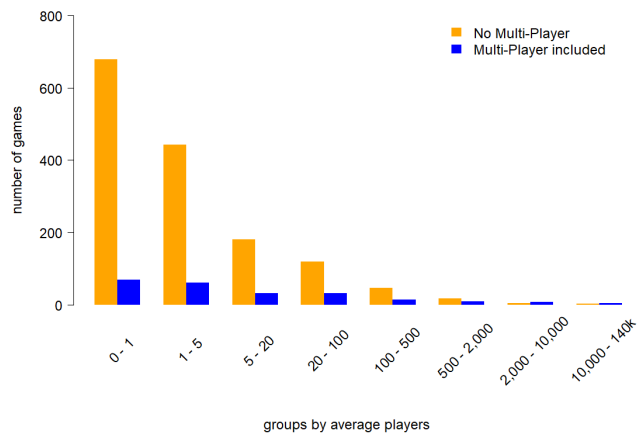


Figure 6.11: Multi-Player

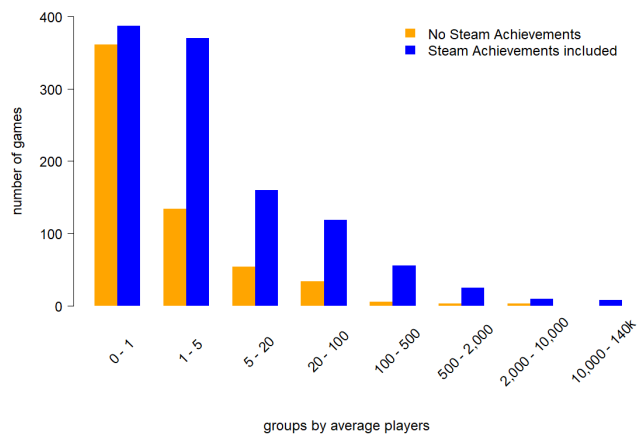


Figure 6.12: Steam Achievements

Some games are believed to be sold only because they offer Steam Trading Cards. Figure 6.13 shows that they are more often present in the more successful games.

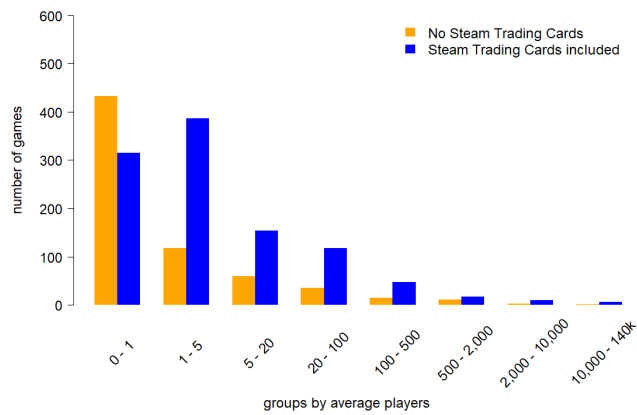


Figure 6.13: Steam Trading Cards

6.3 Developers and Their Experience

Galyonkin (Figure 6.14) shows that after a developer releases at least one game, their next games sell better on average. Games are divided into seven groups based on how many games their developer released in the past in addition to the new game. Sales from this game are used to calculate the average and median sales for all developers over an unspecified period of time.

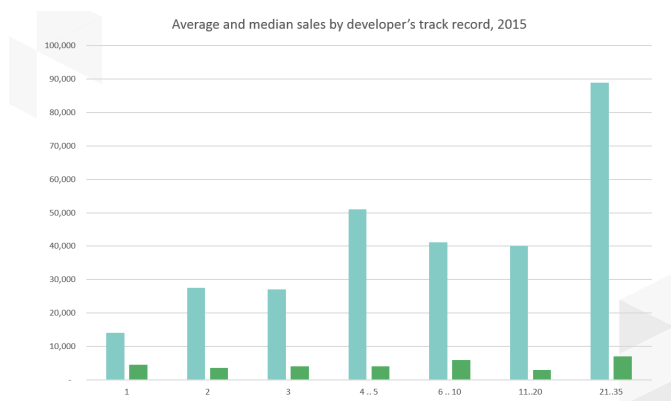


Figure 6.14: Developer track record (Source: post [16])

We attempted to recreate this graph with data about average concurrent players from the first two months after a game's release, presented in Figure 6.15.

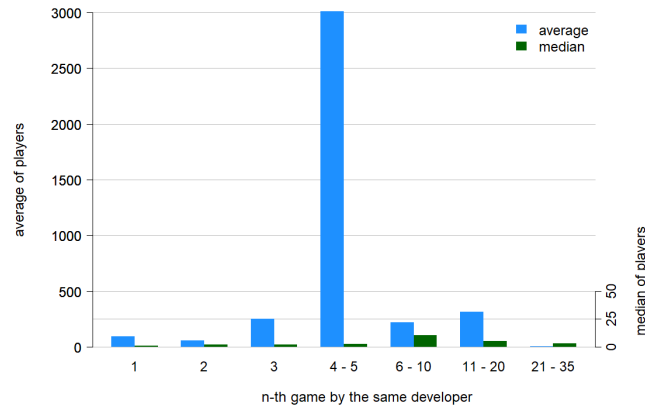


Figure 6.15: Developer track record

The average and median, however, vary too much to represent them on the same scale. Medians are, therefore, represented on a different scale, as seen on the right axis. There are obvious differences. The average for 4th and 5th games stands out even more. This is caused by the two most successful games on Steam in 2015 according to our data - *Fallout 4* and *Grand Theft Auto V*, both being the 4th games on Steam by their respective developers.

Newcomers score the lowest median of players - 1.1. While the second games have less players on average, the median is higher at 2.05 and never drops below this mark with an increasing number of released games. There is a large disparity in the last group of supposedly very experienced developers. However, only seven games are in this group in our dataset – from developers trying to release as many games as possible, likely using assets from their previous games. However, experienced developers are generally more successful in conclusion.

7 Experiments

This chapter describes how the data is split into training, validation, and test set. Preprocessing dependent on these splits is then performed. Namely, text descriptions are transformed into a document-term matrix and missing values are imputed. Once the data is prepared, the prediction of the average number of concurrent players is performed using various techniques and the results are presented.

7.1 Preprocessing

As explained in Section 6.1, the way games were allowed on Steam changed at the end of August, 2013. This was a significant change to the way games were released. Thus, only games released since September, 2013 are further processed, 4,267 entries in total.

The dataset is split into training, validation, and test set in the following manner: 60 % training, 20 % validation, 20 % test, where the oldest games are in the training set, games released after that are in the validation set, and the test set consists of the newest games available in the dataset. We avoided performing cross-validation as it would result in predicting old games from new ones in numerous cases. The training set consists of 2,560 examples, validation and test set have 853 examples each.

The splits are further processed individually.

7.1.1 Descriptions

Each game has a short and a long description provided by its developer. Short descriptions are more visible on the store page and could be expected to have more significance in the evaluation process as customers are more likely to read them. However, experiments have shown that including information about the short descriptions has a lesser effect compared to long descriptions. Thus, only the long descriptions are included further.

A document-term matrix is created from all the descriptions in the training set using the R package *tm*. This matrix is stored for later use when the descriptions in the validation and the test set are processed.

Their document-term matrices are built using the terms from the stored matrix to ensure compatibility across the sets.

Information gain is utilized to select the top 50 most significant terms. The number of selected terms was experimentally determined on the validation set. The top 50 include, sorted by importance: new, experience, combat, content, series, all-new, online, improved, includes, players, levels, puzzles, edition, customize, tactical, choose, war, team, definitive, character, events, franchise, including, multi-player, battle, expansive, puzzle, enhanced, powerful, iconic, platformer, roster, build, recruit, mode, classes, epic, system, engine, among, turn-based, overhaul, stunning, returns, playable, version, history, squad, latest, range.

7.1.2 Imputing Missing Values

There are numerous packages for R implementing missing value imputation, such as *mice* or *missForest*. However, these packages perform imputation on a single dataset and simply return a new dataset without missing values. They do not allow applying the same process on new data to ensure that missing values are handled in the same way. We found no simple implementation for this problem and therefore decided to create our own solution based on Random Forests. It consists of two functions:

Learner: This function is designed for training data. It detects attributes with missing values and only keeps complete attributes (with no missing values). The attributes with missing values are then individually processed. A random forest is trained on observations where the value is available. This model is then applied on the rest of the observations. A new model is trained for each attribute with missing values. The function returns the dataset with imputed values as well as the trained models.

Predict: This function is applied on new data and expects a list of previously trained models. It assumes there are no missing values for attributes that were not processed in the learning phase. (This would indicate a problem during the collection of new data). It operates similarly to the learner function, except it uses the models trained previously to ensure all missing values are computed using the same settings.

7.2 Evaluation

This section describes experiments under several scenarios: regression, and binary classification with three different settings. The results are discussed and further experiments, attempting to find a subset of data providing better results, are suggested.

The class attribute "*Players*" is continuous and ranging from 0 to 135,300 but only few observations reach above 10,000 while the majority are below 10. The experiments were adjusted based on these facts. Four scenarios were created for evaluation:

1. Regression – predicting a numeric value. Binary logarithm is applied to the numeric attribute *Players* to level the large disparity between games with a few players and the games on the other side of the spectrum. 1 must be added before applying the logarithm to prevent the resulting numbers from reaching values below zero. The resulting values range from 0 to 17 with median being 1.85 and mean being 2.75. This still preserves the fact that most games struggle to reach a high number of players on average. Simultaneously, two games with 10,000 and 100,000 players on average can both be considered highly successful despite there being a significant numeric difference between them.

Binary class scenarios where the games are divided into two classes based on the original values of *Players*:

2. [0,1], (1,-): The goal is to determine whether a game will even reach a measurable audience.
3. [0,10], (10,-): Games above 10 players on average may sell enough copies to fund their development, depending on the budget.
4. [0,100], (100,-): Games above 100 players on average likely received attention from popular critiques and entertainers boosting their sales.

7.2.1 Regression

Successfully predicting the exact number of average concurrent players is the ideal scenario. Since there are few games with a high number of

players and many games with a low number of players, the training set was oversampled in such a way that the more players a game has, the more times it is duplicated, up to 15 times. The process of oversampling was fine-tuned on the validation set. The following five algorithms were used for this task:

- Linear model - *lm* from the *stats* package (default settings)
- Recursive Partitioning and Regression Tree – *rpart* from the *rpart* package ($cp = 0.005$)
- Random Forest – *randomForest* from the *randomForest* package ($ntree = 500$)
- Gaussian Process – *gausspr* from the *kernlab* package (default settings)
- Support Vector Machines – *svm* from the *e1071* package (kernel = "polynomial", $coef0 = 0.5$, $gamma = 0.003$)

The data splits contain 214 attributes. Chi squared is used to compute the importance of all attributes. Experiments on validation data showed best results after removing attributes with no significance – 28 in total. This removal of non-significant attributes was performed in all other experiments. The top 30 attributes suggest the importance of the following:

- Minimum and maximum of average players across previous games by the same developer and publisher, and Gini index of these numbers for the developer
- GPU and storage requirements
- Tags: Open World, Third-Person, Sandbox, Story-Rich
- Support for Spanish, French, Polish, Italian, Russian, German, Portuguese-Brazil and the total number of supported languages
- Genres: Indie and Casual
- Presence of DRM in any form and EULA
- Long description's length
- Launch price
- Age requirements
- Whether the game is a sequel
- Screenshot's average saturation and number of all distinct colors
- Presence of multi-player

Table 7.1 shows results on the test set. Correlation coefficient and Relative Root Squared Error are provided as well as the percentage of predictions falling within a specified distance from the actual value. Pearson correlation coefficient produces values from -1 to +1, where 1 means a perfect fit. Relative Root Squared Error is calculated as Root Mean Squared Error where the sum of differences between actual and predicted values is divided by the sum of differences between the mean and actual values. 100 % indicates that simply using the mean as the predicted value gives the same results. Values below 100 % indicate that the model provides better results.

Table 7.1: Average players prediction (regression)

Algorithm	Cor.	RRSE	+1	+2	+3
Baseline	-	100 %	59.7 %	82.9 %	92.4 %
Linear model	0.68	75.4 %	51.2 %	79.7 %	91.4 %
RPART	0.58	95.5 %	39.6 %	70.5 %	85.9 %
Random Forest	0.7	71.8 %	51.6 %	81.7 %	92.5 %
Gaussian Process	0.67	74.4 %	49.4 %	79.1 %	91.3 %
SVM	0.7	73.5 %	54.0 %	82.1 %	91.4 %

+1 from actual says how many percent of predictions were off by maximum of 1, either above or below the actual value. The baseline for this is taken as an interval that covers the maximum number of observation. Because of the distribution, where most games are close to 0 players and the higher the number of players, the less games around that number, the baseline interval starts at 0 and ends at 2. This means that the value of 1 is predicted for all observations. This is performed analogously for +2 and +3.

Random Forest scores the best results with 0.7 correlation and 72 % RRSE, closely followed by SVM with 0.7 correlation and 74 % RRSE. Gaussian Process and even a simple linear model give only slightly worse results, while RPART scores significantly below all other algorithms.

Neither Random Forest nor SVM manage to provide better results than baseline when it comes to falling within a specified interval. They are either close to baseline or, in the case of the strictest interval +1, below it.

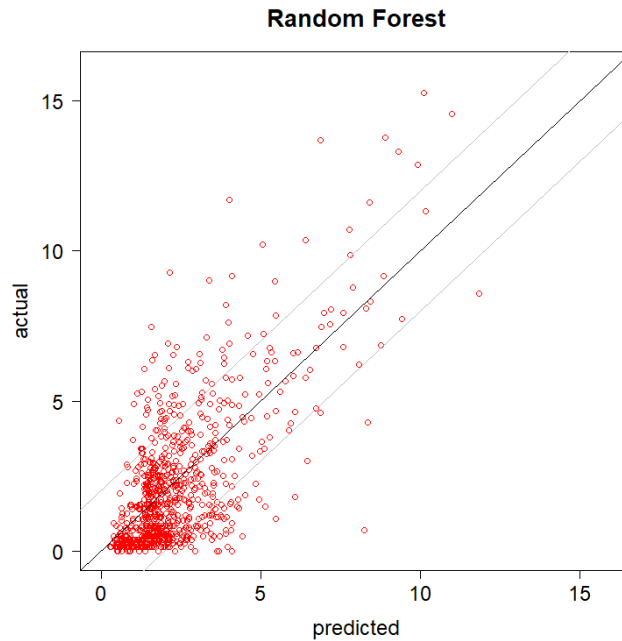


Figure 7.1: Relation between actual and predicted values for Random Forest

Figure 7.1 shows the relation between predicted and actual values for Random Forest and Figure 7.2 for SVM. Ideally, all points should be located on the black diagonal line or at least between the grey lines indicating a deviation by 2 from the actual value. We concluded that deviation by ± 2 still provides a precise estimate of a game's success.

Both algorithms have trouble with under-estimating games with higher actual number of average concurrent players. This is an understandable error as games may gain popularity by investing into promotions around release, attracting the interest of content creators after release etc., which is impossible to capture by the used data. Random Forest has higher tendency to over-estimate games with an actually low number of average concurrent players. We consider this a bigger issue as it may give a false hope to a developer who ends up creating a game not enough players will buy, possibly resulting in not having its development even funded. SVM may, therefore, be

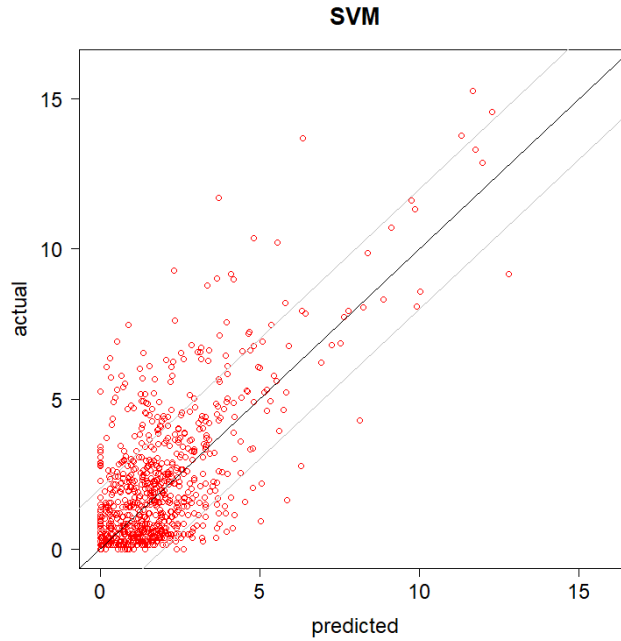


Figure 7.2: Relation between actual and predicted values for SVM

considered better than Random Forest for this task. However, neither successfully cover enough observations to call their results satisfactory.

We also attempted not to use our missing value imputation method described in Subsection 7.1.2. Random Forest was used in this experiment as it implements its own handling of missing values. The results are only marginally worse compared to the results in Table 7.1, substantially differing in covering the interval ± 1 from actual, where we saw a drop from 51.6 % to 48.3 %. 80.8 % of predictions fall within ± 2 from actual, compared to 81.7 % previously. RRSE reached 72.23 % and the remaining metrics did not change.

7.2.2 Classification

Next, we focus on binary classification. The task is to detect games that reached more concurrent players on average than a certain threshold. This is performed in three settings depending on the threshold where games are split (1, 10, and 100 players). All examples are present in the

datasets across these experiments. Oversampling and undersampling showed no improvement on the validation set in these scenarios. Five algorithms were used:

- Recursive Partitioning and Regression Tree – *rpart* from the *rpart* package ($cp = 0.005$)
- General linear model - *glm* from the *stats* package ($family = binomial(link = probit)$)
- Random Forest – *randomForest* from the *randomForest* package ($ntree = 500$)
- Support Vector Machines – *svm* from the *e1071* package ($kernel = "polynomial", coef0 = 0.5, gamma = 0.003$)
- Naïve Bayes – *naiveBayes* from the *e1071* package (default settings)

Results for the scenario of detecting games with more than 1 player on average among all games are presented in Table 7.2. While GLM, NB, and SVM score a high recall, their precision falls around 50 %. Random Forest has a balanced precision and recall – both at 67 % – and its accuracy is at 73 %, 13 % above baseline. A confusion matrix for Random Forest can be seen in Table 7.3. RPART scores similar accuracy and the best precision at 73 % at the cost of the lowest recall at 50 %. This task proves to be difficult, some of which may be attributed to the fact that there is only a small difference between games with 0.9 and 1.1 players on average.

Table 7.2: Predicting games with more than 1 player on average; baseline accuracy 59.3 %

Algorithm	Accuracy	Precision	Recall	F1
RPART	72.3 %	73.4 %	50.1 %	59.6 %
GLM	60.8 %	51.0 %	91.9 %	65.6 %
Random Forest	73.2 %	67.0 %	67.1 %	67.1 %
SVM	63.7 %	53.3 %	86.5 %	65.9 %
Naïve Bayes	54.9 %	47.0 %	86.7 %	61.0 %

The next scenario is detecting games with more than 10 players on average. The results can be seen in Table 7.4. SVM stands out with the highest precision of 88 % but also the lowest recall at 29 %.

Table 7.3: Predicting games with more than 1 player on average – confusion matrix for Random Forest

	Reference	
Prediction	≤ 1	> 1
≤ 1	233	115
> 1	114	391

Random Forest offers recall of almost 40 % at the cost of somewhat lower precision at 81 %. Its confusion matrix is presented by Table 7.5. Overall, games with more than 10 players on average can be detected with a relatively high confidence but only for a portion of them.

Table 7.4: Predicting games with more than 10 player on average; baseline accuracy 79.1 %

Algorithm	Accuracy	Precision	Recall	F1
RPART	83.5 %	66.4 %	42.1 %	51.5 %
GLM	85.0 %	82.9 %	35.4 %	49.6 %
Random Forest	85.5 %	81.4 %	39.3 %	53.0 %
SVM	84.3 %	87.9 %	28.7 %	43.2 %
Naïve Bayes	80.3 %	53.4 %	44.4 %	48.5 %

Table 7.5: Predicting games with more than 10 player on average – confusion matrix for Random Forest

	Reference	
Prediction	≤ 10	> 10
≤ 10	659	108
> 10	16	70

Games with more than 100 players on average were detected next and Table 7.6 shows the results. This task is difficult since 95 % games have no more than 100 players on average. Still, all classifiers except Naïve Bayes managed to reach an accuracy higher than 96 %. SVM scored the highest precision at 86 %, followed by Random Forest with 85 %. Random Forest reached higher recall at 51 % compared to 44 % for SVM and its confusion matrix can be seen in Table 7.7.

Table 7.6: Predicting games with more than 100 player on average; baseline accuracy 95.0 %

Algorithm	Accuracy	Precision	Recall	F1
RPART	96.1 %	63.2 %	55.8 %	59.3 %
GLM	96.6 %	76.9 %	46.5 %	58.0 %
Random Forest	97.1 %	84.6 %	51.2 %	63.8 %
SVM	96.8 %	86.4 %	44.2 %	58.5 %
Naïve Bayes	89.7 %	27.7 %	65.1 %	38.9 %

Table 7.7: Predicting games with more than 100 player on average – confusion matrix for Random Forest

	Reference	
Prediction	≤ 100	> 100
≤ 100	806	21
> 100	4	22

In conclusion, regression showed a strong correlation between core game features and the average number of concurrent players. Detecting the more successful games is possible with relatively high precision but low recall. In general, high precision is preferred to avoid falsely marking a game as potentially highly successful. A high baseline proved to be the hurdle in the conducted experiments as games with nearly no players on average form a substantial part of the dataset and cannot be reliably separated from the rest of the games. Therefore, we attempted to find a subset with lower baseline, providing more reliable results as described in the following section.

7.2.3 Experiments on Subsets

Since precision is preferred to recall when it comes to determining how well a game will be received, the next experiment focused on finding a subset of games for which predictions would be of higher quality. This section describes the process of finding such a subset and results from experiments on these subsets are presented.

The obvious way of finding an easily identifiable subset of data is by taking only observations with a certain value of one of its attributes,

e.g. only games supporting multi-player. We compared regression results on various subsets such as single-player games, multi-player games, games from a certain price range, games offering Steam Trading Cards etc. However, the only attributes that provided a noteworthy improvement were the number of previous games by both the developer and the publisher. This aligns with the findings in Section 6.3 where we showed that consecutive games by the same developer generally have more players. Thus, the data was reduced according to these attributes.

Intuitively, after releasing one game, the second game can be expected to have a similar or higher number of average players as the previous one as they likely appeal to the same audience and this audience is already invested in the previous title.

Apart from limiting the data to games from developers/publishers who previously made one game, we also examined using 2, 3, and 5 as limits on how many previous games were made by them. Limit to one game meant that validation and test sets only contained games from developers/publishers whose at least one game was present in the training set. The training set was not modified. The new size of the validation set was 388 and 373 for the test set.

Results for the same regression models as used previously can be seen in Table 7.8. SVM and Gaussian process saw an improvement of 0.07 in correlation and 7 % in RRSE compared to the previous results. However, despite the baseline for intervals being lower, the models, again, struggle to reach it.

Table 7.8: Average players prediction (regression, limit of 1 for developer/publisher)

Algorithm	Cor.	RRSE	+1	+2	+3
Baseline	-	100 %	53.1 %	77.2 %	89.3 %
Linear model	0.75	68.5 %	49.3 %	78.0 %	90.1 %
RPART	0.61	91.9 %	34.3 %	64.9 %	82.6 %
Random Forest	0.72	69.3 %	48.3 %	77.5 %	89.0 %
Gaussian Process	0.74	67.0 %	47.2 %	77.7 %	90.3 %
SVM	0.77	65.9 %	49.3 %	81.0 %	90.6 %

Next, we set the limit to 2. Validation and test sets were reduced to games whose developer/publisher had at least two games in the training set, 297 and 278 games, respectively. Consequently, games by developers and publishers with only one record in the training set were removed, leaving 1,424 examples in this set.

In results presented by Table 7.9, SVM stands out as the best-performing algorithm, scoring a high correlation of 0.82 and RRSE of 57 %. Even though the baseline still covers a large percentage of the examples, SVM covers significantly more examples. The relation between the actual and predicted values for SVM is shown in Figure 7.3. While there are games outside the grey lines, only a few are located significantly far. In fact, the games whose actual number of players was high were usually predicted with an acceptable deviation.

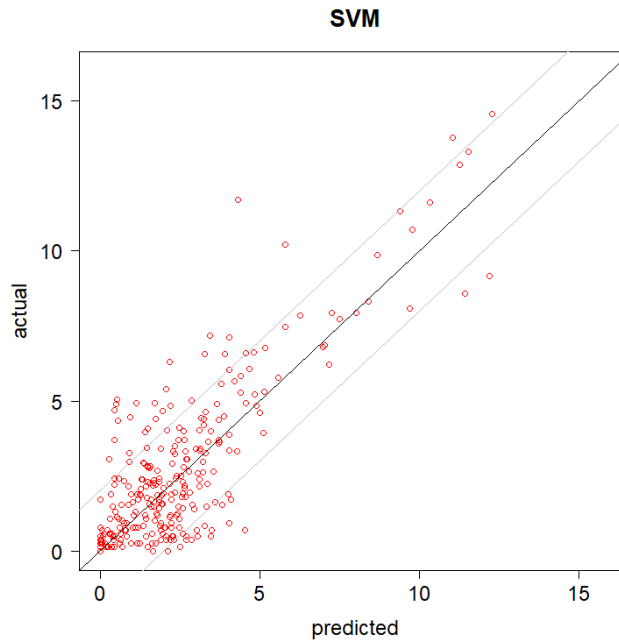


Figure 7.3: Relation between actual and predicted values for SVM

Analogously to the limit of 2, limits of 3 and 5 were applied and the corresponding results are shown in Table 7.10 and Table 7.11. The improvements in correlation and RRSE are only marginal and while the limit of 2 meant that 33 % of the original test set was covered, only

Table 7.9: Average players prediction (regression, limit of 2 for developer/publisher)

Algorithm	Cor.	RRSE	+1	+2	+3
Baseline	-	100 %	51.1 %	76.6 %	88.5 %
Linear model	0.79	61.9 %	53.2 %	77.7 %	90.6 %
RPART	0.68	81.8 %	43.9 %	70.5 %	83.5 %
Random Forest	0.76	65.4 %	52.5 %	78.1 %	90.6 %
Gaussian Process	0.80	60.7 %	53.2 %	77.7 %	92.1 %
SVM	0.82	57.5 %	55.8 %	83.5 %	93.5 %

28 % and 22 % were covered with the limits of 3 and 5, respectively. Thus, the subset of games by developers/publishers who previously released at least two games was chosen as the most suitable one. This, again, aligns with findings in Section 6.3.

Table 7.10: Average players prediction (regression, limit of 3 for developer/publisher)

Algorithm	Cor.	RRSE	+1	+2	+3
Baseline	-	100 %	50.0 %	74.6 %	86.7 %
Linear model	0.80	60.6 %	50.0 %	75.8 %	88.8 %
RPART	0.71	77.2 %	42.1 %	69.2 %	84.2 %
Random Forest	0.77	64.3 %	51.7 %	75.8 %	90.0 %
Gaussian Process	0.80	60.1 %	50.4 %	77.5 %	90.8 %
SVM	0.83	56.9 %	52.5 %	79.6 %	92.1 %

Table 7.11: Average players prediction (regression, limit of 5 for developer/publisher)

Algorithm	Cor.	RRSE	+1	+2	+3
Baseline	-	100 %	54.8 %	73.9 %	85.6 %
Linear model	0.83	57.0 %	50.0 %	77.7 %	90.4 %
RPART	0.60	86.6 %	30.9 %	66.0 %	79.8 %
Random Forest	0.78	63.5 %	50.0 %	77.1 %	89.9 %
Gaussian Process	0.81	59.0 %	47.3 %	75.0 %	90.8 %
SVM	0.84	54.4 %	55.3 %	76.6 %	92.6 %

The same binary classification experiments with the same settings were performed with the limit of 2 previous games by developer/publisher. First, games with more than 1 player on average were detected and the results can be seen in Table 7.12. Overall, there was a large increase in precision. Previously best model, Random Forest went up from 67 % to 81 % in precision and from 67 % to 84 % in recall (its confusion matrix is presented in Table 7.13). Naïve Bayes reached the highest precision of 89 % at the cost of the lowest recall at 37 %.

Table 7.12: Predicting games with more than 1 player on average; baseline accuracy 69.1 % (limit of 2 for developer/publisher)

Algorithm	Accuracy	Precision	Recall	F1
RPART	73.7 %	80.5 %	81.8 %	81.1 %
GLM	64.7 %	83.6 %	60.9 %	70.5 %
Random Forest	75.5 %	81.3 %	83.9 %	82.6 %
SVM	67.3 %	82.2 %	67.2 %	73.9 %
Naïve Bayes	53.2 %	88.8 %	37.0 %	52.2 %

Table 7.13: Predicting games with more than 1 player on average (limit of 2 for developer/publisher) – confusion matrix for Random Forest

	Reference	
Prediction	≤ 1	> 1
≤ 1	49	31
> 1	37	161

Games with more than 10 players on average were detected in the next step. Table 7.14 shows that SVM saw a large improvement to 95 % precision and 45 % recall and was, overall, the best-performing classifier. Table 7.15 shows its confusion matrix.

Lastly, games with over 100 players were detected. Despite the baseline still being very high 91.4 %, Random Forest scored outstanding results of 97.5 % accuracy, 95 % precision and 75 % recall, as presented in Table 7.16. Its confusion matrix presented by Table 7.17 shows only one example falsely classified as more successful.

Table 7.14: Predicting games with more than 10 player on average; baseline accuracy 71.9 % (limit of 2 for developer/publisher)

Algorithm	Accuracy	Precision	Recall	F1
RPART	77.0 %	60.3 %	52.6 %	56.2 %
GLM	82.7 %	77.8 %	53.8 %	63.6 %
Random Forest	83.1 %	78.2 %	55.1 %	64.7 %
SVM	83.8 %	94.6 %	44.9 %	60.9 %
Naïve Bayes	80.2 %	79.5 %	39.7 %	53.0 %

Table 7.15: Predicting games with more than 10 player on average (limit of 2 for developer/publisher) – confusion matrix for SVM

	Reference	
Prediction	≤ 10	> 10
≤ 10	198	43
> 10	2	35

In conclusion, limiting the dataset to games from experienced developers/publishers allows reliably detecting the more successful games.

Table 7.16: Predicting games with more than 100 player on average; baseline accuracy 91.4 % (limit of 2 for developer/publisher)

Algorithm	Accuracy	Precision	Recall	F1
RPART	96.0 %	76.0 %	79.2 %	77.6 %
GLM	96.4 %	79.2 %	79.2 %	79.2 %
Random Forest	97.5 %	94.7 %	75.0 %	83.7 %
SVM	96.4 %	93.8 %	62.5 %	75.0 %
Naïve Bayes	93.9 %	60.6 %	83.3 %	70.2 %

Table 7.17: Predicting games with more than 100 player on average (limit of 2 for developer/publisher) – confusion matrix for Random Forest

	Reference	
Prediction	≤ 100	> 100
≤ 100	253	6
> 100	1	18

7.3 Résumé

While the experiments showed a significant correlation between core game features and the number of average players after release on the whole dataset, the results are often close to baseline. The quality of predictions is higher for some games while lower for others. Therefore, we found a subset of games covering 33 % of the test data by limiting it to games from developers/publishers who previously released at least two games. Games in this subset can be predicted more reliably. It is possible to directly predict the number of players using SVM, reaching correlation of 0.82. It is also possible to detect the most successful games (with more than 100 players on average) with 95 % precision using Random Forest.

7.4 Implementation

An end-user application was developed to provide predictions for unreleased games. It uses the SVM model for regression trained on a subset of games from developers/publishers with at least 2 games on their record. The application is written in the web application framework Shiny¹ and therefore requires R.

It consists of a form where a user is requested to input information as if they were planning to release a game on Steam. If less than two games from the entered developer and publisher are found, the application informs the user about this fact and consequently not being able to provide any reliable predictions.

1. <https://shiny.rstudio.com/>

GamesPred

GamesPred allows you to insert info about your upcoming (even unannounced!) game and it will attempt to predict its average number of concurrent players in the first two months following the game's release.

Note: Not all games can be reliably predicted. You will be immediately notified when that is the case

*fields are required, the rest can be left empty but at the cost of decreased accuracy of predictions

Name

Developer*

Publisher*

Age Requirements

Release date*

Price (USD)

Short Description

Looking good!

Hit "OK" in the bottom when ready. Your prediction will show up here.

Predicted average number of concurrent players in the first 2 months after release: **25 - 422** (confidence 80 %)

Games which got a similar prediction in the past:

Name	Actual number of players	Prediction	Features that these games share with your game
TransOcean 2: Rivals	103	24 - 399	Windows, Mac, Single-Player, Multi-Player, Steam Achievements, Steam Trading Cards, Required RAM, Third-party DRM, Third-party Eula, Weekday of release, English, German
The Banner Saga 2	220	24 - 393	Windows, Mac, Single-Player, Steam Achievements, Steam Trading Cards, Controller Support, Required GPU, Number of Trailers, Weekday of release, English
Dead Island: Riptide Definitive Edition	74	22 - 370	Windows, Linux, Single-Player, Multi-Player, Co-op, Steam Achievements, Steam Trading Cards, Controller Support, Required RAM, Required DirectX version, Third-party DRM, Third-party Eula, TagOpenWorld, Weekday of release, English, German, Action
Sherlock Holmes: The Devil's Daughter	108	29 - 485	Age Requirements, Windows, Single-Player, Steam Achievements, Steam Trading Cards, Controller Support, Required DirectX version, TagOpenWorld, English, German, Adventure, Action
Hard Reset Redux	49	20 - 330	Windows, Single-Player, Steam Achievements, Steam Trading Cards, Controller Support, TagFPS, English, German, Adventure, Action
The Technomancer	210	37 - 614	Windows, Single-Player, Steam Achievements, Steam Trading Cards, Controller Support, Required RAM, Required DirectX version, Third-party DRM, Third-party Eula, TagOpenWorld, Weekday of release, English, German, Action

Figure 7.4: Application interface

The input is processed in a similar way as described in Chapter 5. The output consists of an estimated number of average concurrent players in the first two months after release. In addition, 10 games with the most similar predictions are displayed, along with their actual number of players for comparison, and the features these games share with the game newly entered by the user. The interface is shown in Figure 7.4.

The application is publicly available at:
<https://github.com/trnenym/GamesPred> under the GPLv3 license².

2. <https://www.gnu.org/licenses/gpl-3.0.en.html>

8 Conclusion

The research described in this thesis showed that while the PC games market has been growing, there are no detailed studies on the topic of predicting the success on this market. We discussed factors influencing games' success after release and suggested estimating this success long before release.

For this purpose, dataset containing information about nearly 10,000 games was created, with over 4,600 of them being suitable for evaluation. This data already provided an insight into what may influence a game's success, such as detailed genre or presence of certain features.

We discovered a high correlation between core features known before release and the average number of concurrent players in the first two months after release. Having experience as a developer or publisher turned out to be the most determining factor in prediction accuracy. When limiting data to games from developers or publishers with at least two games on their record, we were able to reach 0.82 correlation using Support Vector Machines when directly predicting the average number of players. Games reaching over 100 players on average can be detected with 95% precision and 75% recall using Random Forest. This covers 33 % of the data.

There is a publicly available application which allows developers to input information about their upcoming game and provides them with a prediction along with a comparison to other games.

Possible future extensions include:

- updating the dataset with new games, seeing how the market has developed, and the impact on predictions of newest games
- discovering a way to reliably predict a larger subset of games
- utilizing other machine learning methods such as clustering
- studying and potentially predicting trends (such as a popular genre)

Bibliography

1. NOMIKOS, Petros M. (ed.). *2016 IEEE Conference on Computational Intelligence and Games*. 2016 IEEE Conference on Computational Intelligence and Games. 2016.
2. SIWEK, STEPHEN E. *Video Games in the 21st Century: The 2017 Report* [online]. 2017 [visited on 2017-04-12]. Available from: http://www.theesa.com/wp-content/uploads/2017/02/ESA_EconomicImpactReport_Design_V3.pdf.
3. MCCOURT, J. *Year-end DEG Home Entertainment Spending* [online]. 2016 [visited on 2017-04-12]. Available from: http://degonline.org/wp-content/uploads/2016/01/External_2015-Year-end-DEG-Home-Entertainment-Spending-1-5-2016.pdf.
4. *Yearly Box Office* [online]. 2017 [visited on 2017-04-12]. Available from: <http://www.boxofficemojo.com/yearly/>.
5. VALVE. *Valve Launches Steam Greenlight* [online]. 2012 [visited on 2017-04-18]. Available from: <http://store.steampowered.com/news/8761/>.
6. VALVE. *Steam Search* [online]. 2017 [visited on 2017-01-06]. Available from: <http://store.steampowered.com/search/?category1=998>.
7. SIFA, Rafet; SRIKANTHY, Sridev; DRACHENZ, Anders; OJEDA, Cesar; BAUCKHAGE, Christian. Predicting Retention in Sandbox Games with Tensor Factorization-based Representation Learning. In: NOMIKOS, Petros M. (ed.). *2016 IEEE Conference on Computational Intelligence and Games*. 2016, p. 142.
8. TAMASSIA, Marco; RAFFEY, William; SIFAZ, Rafet; DRACHENX, Anders; ZAMBETTA, Fabio; HITCHENS, Michael. Predicting Player Churn in Destiny: A Hidden Markov Models Approach to Predicting Player Departure in a Major Online Game. In: NOMIKOS, Petros M. (ed.). *2016 IEEE Conference on Computational Intelligence and Games*. 2016, p. 325.
9. SHAKER, Noor; ABOU-ZLEIKHA, Mohamed. Transfer Learning for Cross-Game Prediction of Player Experience. In: NOMIKOS, Petros M. (ed.). *2016 IEEE Conference on Computational Intelligence and Games*. 2016, p. 209.

BIBLIOGRAPHY

10. BEAUJON, Walter Steven. Predicting Video Game Sales in the European Market. 2012. Available also from: https://www.few.vu.nl/nl/Images/werkstuk-beaujon_tcm243-264134.pdf.
11. EHRENFELD, Steven Emil. *Predicting Video Game Sales Using an Analysis of Internet Message Board Discussions*. 2011. Available also from: https://sdsu-dspace.calstate.edu/bitstream/handle/10211.10/1073/Ehrenfeld_Steven.pdf. Master's thesis. San Diego State University.
12. GHATTAMANENI, Sriram; KOMARRAJU, Agastya kumar. The Game Prophet: Predicting the success of Video Games. 2012. Available also from: https://cepd.okstate.edu/files/Analytics_Ghatta.pdf.
13. *VGChartz Methodology* [online]. 2017 [visited on 2017-03-25]. Available from: <http://www.vgchartz.com/methodology.php>.
14. GETOMER, James; OKIMOTO, Michael; CLEAVER, Jennifer. Understanding the Modern Gamer. 2012. Available also from: https://ssl.gstatic.com/think/docs/understanding-the-modern-gamer_research-studies.pdf.
15. ORLAND, Kyle. *Steam Gauge: Do strong reviews lead to stronger sales on Steam?* [online]. 2014 [visited on 2017-03-21]. Available from: <https://arstechnica.com/gaming/2014/04/steam-gauge-do-strong-reviews-lead-to-stronger-sales-on-steam/>.
16. GALYONKIN, Sergey. *On Indiepocalypse: What is really killing indie games* [online]. 2015 [visited on 2017-03-21]. Available from: <https://galyonk.in/on-indiepocalypse-what-is-really-killing-indie-games-3da3c3a1ea76>.
17. *About - Twitch* [online]. 2017 [visited on 2017-03-21]. Available from: <https://www.twitch.tv/p/about>.
18. HERNANDEZ, Danny. *Game Creator Success on Twitch: Hard Numbers* [online]. 2016 [visited on 2017-03-21]. Available from: <https://blog.twitch.tv/https-blog-twitch-tv-game-creator-success-on-twitch-hard-numbers-688154815817>.
19. *Faeria - Steam Charts* [online]. 2017 [visited on 2017-03-13]. Available from: <https://imgur.com/1FE9KpJ>.

BIBLIOGRAPHY

20. GALYONKIN, Sergey. *About - SteamSpy - All the data and stats about Steam games* [online] [visited on 2017-03-12]. Available from: <https://steamspy.com/about>.
21. GRAY, James. *About - Steam Charts* [online] [visited on 2017-03-12]. Available from: <http://steamcharts.com/about>.
22. *Steam Database* [online] [visited on 2017-03-12]. Available from: <https://steamdb.info/>.
23. *User:RJackson/StorefrontAPI - Official TF2 Wiki* [online]. 2013 [visited on 2017-03-12]. Available from: <https://wiki.teamfortress.com/wiki/User:RJackson/StorefrontAPI#appdetails>.
24. BBC. *Copyright row dogs Spore release* [online]. 2008 [visited on 2017-03-12]. Available from: <http://news.bbc.co.uk/2/hi/technology/7604405.stm>.
25. VALVE. *Introducing Early Access* [online] [visited on 2017-03-12]. Available from: <http://store.steampowered.com/earlyaccessfaq/>.
26. *DirectX Graphics and Gaming* [online] [visited on 2017-03-12]. Available from: <https://msdn.microsoft.com/en-us/library/windows/desktop/ee663274%28v=vs.85%29.aspx>.
27. *The Signal From Tlva - SteamSpy* [online] [visited on 2017-04-25]. Available from: <https://steamspy.com/app/457760>.
28. *Steam Workshop :: August 28th Batch of Greenlit Titles* [online]. 2013 [visited on 2017-04-25]. Available from: <https://steamcommunity.com/sharedfiles/filedetails/?id=171260210>.